

Definition. Let G be a set. A *binary operation* on G is a function that assigns each ordered pair of elements of G an element G .

Definition. Let G be a set together with a binary operation. We say G is a *group* under this operation if the following three properties hold:

1.

2.

3.

GROUPS	NOT GROUPS

Some Properties of Groups:

1. The identity in a group is unique.

2. Left and right cancellation laws hold.

3. The inverse of any element is unique.

Theorem (Socks and Shoes). For group elements a and b , $(ab)^{-1} = b^{-1}a^{-1}$.